PSEUDO-COHERENCE IS PRESERVED BY THE DERIVED TENSOR PRODUCT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

Editorial note. The citation fields (source_id, source_version, authors, source_url, license.name, source_title) are identical to those of every other card this lane has produced from this chapter, so the batch selector treats them as one source.

Allowed standard prerequisites (editorial).

- Pseudo-coherence of a complex at a degree, defined via a bounded complex of finite free modules mapping to it. defined earlier in the same section, source lines 16360-16485. Supplies the hypothesis on both complexes and the conclusion to be proved.
- The tensor product of the comparison maps induces the stated isomorphisms and surjection on cohomology. used at proof lines 16500-16510. Is the statement that has to be verified, by the spectral sequence of tors as the author notes.
- Spectral sequence of tors for a total tensor product of complexes. the author refers to it at proof line 16510 as the tool for the verification. Supplies the mechanism for the cohomology comparison.

Source reference. Original lemma 0DJE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16974–16974. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 064U: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7878–7878. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring. Let K, L be objects of $D(R)$.*

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(1) *If K is n-pseudo-coherent and $H^i(K) = 0$ for $i > a$ and L is m-pseudo-coherent and $H^j(L) = 0$ for $j > b$, then $K \otimes^{\mathbf{L}}_R L$ is t-pseudo-coherent with $t = \max(m + a, n + b)$.*
(2) *If K and L are pseudo-coherent, then $K \otimes^{\mathbf{L}}_R L$ is pseudo-coherent.*

Proof. Proof of (1). We may assume there exist bounded complexes $K^\bullet$ and $L^\bullet$ of finite free R-modules and maps $\alpha : K^\bullet \to K$ and $\beta : L^\bullet \to L$ with $H^i(\alpha)$ and isomorphism for $i > n$ and surjective for $i = n$ and with $H^i(\beta)$ and isomorphism for $i > m$ and surjective for $i = m$. Then the map

$$\alpha \otimes^{\mathbf{L}} \beta : \mathrm{Tot}(K^\bullet \otimes_R L^\bullet) \to K \otimes^{\mathbf{L}}_R L$$

induces isomorphisms on cohomology in degree i for $i > t$ and a surjection for $i = t$. This follows from the spectral sequence of tors (details omitted). Part (2) follows from part (1) and Lemma 064U. □